

URITECT

Bayesian UTI

Literature-Derived Parameter Review Form

Version 1.2

Review panel	Two registered medical technologists
Reviewer copy	Mark MT-01 or MT-02 in Section 10
Model reviewed	uti_bayesian_lr_v1_1_20260926
What is reviewed	Candidate numerical parameters and the rules used to apply them
What is not claimed	Clinical accuracy, local calibration, diagnosis, or treatment validation

Please review the evidence summary first, then complete every item in Part A and Part B independently. Use the comment spaces whenever you recommend a change.

Prepared for expert content review of a thesis screening model

Model reviewed: uti_bayesian_lr_v1_1_20260926 Reviewers for this form: Two registered medical technologists
 Separate final-app validation: Ten registered medical technologists Purpose: Expert content review of proposed literature-derived parameters and their implementation

1. Purpose of This Review

URIRECT uses urine-strip findings and selected urinary symptoms to produce a provisional research estimate for uncomplicated lower urinary tract infection (UTI). This form asks you to review two separate matters:

- Whether each proposed literature-derived value is relevant and acceptable as a candidate parameter for the research model.
- Whether URIRECT's rules for applying those values are understandable and appropriate for the stated research purpose.

This is an expert content review. It does not prove that the final percentage is clinically accurate, locally calibrated, or diagnostic. It does not ask the reviewer to recommend antibiotics or treatment.

2. Short Explanation of the Terms

Term	Plain-language meaning
Likelihood ratio (LR)	A number used to adjust how strongly the available evidence supports UTI.
LR greater than 1	The finding increases support for UTI.
LR less than 1	The finding decreases support for UTI.
Prior probability	The starting assumption before the current strip and symptom findings are applied.
Literature-derived candidate	A value taken from published evidence and proposed for research use; it is not yet locally calibrated.
No update / neutral	The finding does not change the calculation; mathematically, LR = 1.00.

3. Intended Use and Limits

The calculation is limited to symptomatic, nonpregnant women aged 18 to 64 without a urinary catheter, known urinary tract abnormality, immunocompromise, or systemic warning findings. Vaginal discharge or irritation activates a separate consultation pathway instead of the ordinary calculation.

For an eligible calculation, the app shows the provisional percentage, prior, and applied evidence factors. It does not present the number as a diagnosis or as a Low, Moderate, or High UTI category. Culture-confirmed patient-level validation is required before any claim of clinical calibration.

4. How to Rate Each Statement

Please place one check mark in each row.

Rating	Meaning
4	Highly relevant and acceptable as written
3	Relevant, but a minor revision is suggested
2	Major revision is needed
1	Not relevant or not acceptable for the stated research purpose

Use the comment column whenever you select 1, 2, or 3, or when a threshold, source, or wording should be changed.

5. Evidence Summary

Finding or starting assumption	Candidate value	Main source used by URIRECT	How URIRECT uses it
Nitrite positive	LR 5.50	Kurotschka et al. (2024), Table 4	Used when nitrite is positive without the stronger grouped pattern
Leukocyte esterase positive	LR 1.40	Kurotschka et al. (2024), Table 4	Used when source-threshold leukocytes are the strongest dipstick evidence
Dipstick blood positive	LR 1.70	Kurotschka et al. (2024), Table 4	Used when source-threshold blood is the strongest dipstick evidence
Nitrite positive plus leukocytes or blood positive	LR 7.20	Little et al. (2006)	Used once as a grouped dipstick pattern
Nitrite, leukocytes, and blood all confirmed negative	LR 0.22	Little et al. (2006)	Used once as the grouped negative pattern
Dysuria present	LR 1.30	Kurotschka et al. (2024), Table 4	Used when it is the strongest matching symptom rule
Urgency present	LR 1.20	Kurotschka et al. (2024), Table 4	Used when no stronger symptom rule matches
Frequency present	LR 1.10	Kurotschka et al. (2024), Table 4	Used when no stronger symptom rule matches
Dysuria and urgency both present	LR 1.50	Kurotschka et al. (2024), Table 4	Used once instead of multiplying the individual symptom LRs
Starting prior for the narrowly gated population	50%	Bent et al. (2002); Kurotschka et al. (2024)	Provisional starting assumption; not locally calibrated

Little et al. studied 427 women with suspected UTI and used laboratory diagnosis based on European urinalysis guidance. Its dipstick definitions included leukocyte esterase at + or greater and blood at haemolysed trace or greater. The 50% starting value is limited to a study-like population of symptomatic women; it is not a general-population prevalence.

6. Part A - Evidence and Parameter Review

Please rate whether each value is relevant and acceptable as a literature-derived candidate parameter in the stated URITECT research model.

ID	Statement	1	2	3	4	Comment
P1	LR 5.50 for a positive nitrite result	☐	☐	☐	☐	
P2	LR 1.40 for a positive leukocyte esterase result	☐	☐	☐	☐	
P3	LR 1.70 for a positive dipstick blood result	☐	☐	☐	☐	
P4	LR 7.20 for nitrite positive plus leukocytes or blood positive	☐	☐	☐	☐	
P5	LR 0.22 when nitrite, leukocytes, and blood are all confirmed negative	☐	☐	☐	☐	
P6	LR 1.30 when dysuria is present	☐	☐	☐	☐	
P7	LR 1.20 when urgency is present	☐	☐	☐	☐	
P8	LR 1.10 when urinary frequency is present	☐	☐	☐	☐	
P9	LR 1.50 when dysuria and urgency are both present	☐	☐	☐	☐	
P10	A provisional 50% prior is adequately explained and restricted to the intended study-like population	☐	☐	☐	☐	

Part A Comments

Parameters that should be retained without revision:

Parameters that require revision, with the recommended change and reason:

7. How URITECT Applies the Values

URITECT does not multiply every available number. It selects only one dipstick-pattern LR and one urinary-symptom LR. This is intended to reduce double-counting among related findings.

Dipstick Rules

Order	First matching pattern	LR used
A1	Nitrite positive plus source-threshold leukocytes or blood positive	7.20
A2	Nitrite positive without that grouped pattern	5.50
A3	Source-threshold blood positive	1.70
A4	Source-threshold leukocytes positive	1.40
A5	Nitrite, leukocytes, and blood all confirmed negative	0.22
A6	No source-matched rule	1.00; no update

Symptom Rules

Order	First matching symptom pattern	LR used
B1	Dysuria and urgency both present	1.50
B2	Dysuria present	1.30
B3	Urgency present	1.20
B4	Frequency present	1.10
B5	No weighted symptom is reported	1.00; no update

Exact Category Mapping

URITECT result	Treatment in this candidate model
Nitrite Positive	Positive
Leukocytes Small 70, Moderate 125, or Large 500	Positive; corresponds to + or greater
Leukocytes Trace 15	Neutral; no update
Blood Hemolyzed 10, Small 25, Moderate 80, or Large 200	Positive; haemolysed trace or greater
Blood Non-hemolyzed 10	Neutral; not used as the Little et al. blood threshold
Exact Neg or Negative	Confirmed negative
Missing, invalid, unavailable, or unreliable	Neutral; no update, not treated as negative

Visible hematuria is recorded as consultation context and is not numerically weighted when dipstick blood can contribute. Vaginal discharge or irritation stops the ordinary calculation and routes to consultation.

8. Part B - Implementation and Content Review

ID	Statement	1	2	3	4	Comment
I1	Treating leukocyte Trace 15 as neutral rather than positive is appropriate for the cited threshold	☐	☐	☐	☐	
I2	Treating leukocytes Small 70 or greater as positive is appropriate	☐	☐	☐	☐	
I3	Treating Hemolyzed 10 blood or greater as positive is appropriate for the cited threshold	☐	☐	☐	☐	
I4	Treating Non-hemolyzed 10 blood as neutral is appropriate for this evidence mapping	☐	☐	☐	☐	
I5	Missing, invalid, unavailable, and unreliable results should receive no update and should not be treated as negative	☐	☐	☐	☐	
I6	Using only one mutually exclusive dipstick LR is appropriate for reducing duplicate counting	☐	☐	☐	☐	
I7	Using the combined dysuria-and-urgency LR instead of multiplying their individual LRs is appropriate	☐	☐	☐	☐	
I8	Recording visible hematuria as consultation context without adding another numerical LR is appropriate	☐	☐	☐	☐	
I9	Routing vaginal discharge or irritation to consultation instead of multiplying a numerical factor is appropriate	☐	☐	☐	☐	
I10	The intended-population gate and the handling of unknown information are understandable and appropriate	☐	☐	☐	☐	
I11	The model's non-diagnostic and non-calibrated limitations are stated clearly	☐	☐	☐	☐	
I12	The proposed workflow is understandable and feasible for the stated offline screening research purpose	☐	☐	☐	☐	

Part B Comments

Rules or mappings that should be changed, with the recommended change:

9. Overall Expert Decision

Please select one:

- ☐ Accept the proposed literature-derived parameters and implementation as written for the stated thesis research purpose.
- ☐ Accept after the minor revisions described in this form.
- ☐ Major revision and repeat expert review are required.
- ☐ Do not accept for the stated research purpose.

This decision concerns expert content review only. It does not certify clinical diagnostic accuracy, probability calibration, or treatment suitability.

10. Reviewer Information and Declaration

Reviewer detail	Entry
Reviewer code	[] MT-01 [] MT-02
Reviewer name	
Professional title	
PRC registration/license number	
Years of professional practice	
Current institution	

I confirm that I reviewed the proposed literature-derived parameters, cited evidence summary, category mappings, and implementation rules. I understand that this is content review of a thesis screening model and is not clinical validation against urine culture.

Confirmation	Entry
Signature	
Date	

11. For the Researchers Only

Calculate and report the parameter-review and implementation-review results separately. Do not combine this two-medtech parameter review with the planned ten-medtech final-app usability, functionality, or ISO/IEC 25010 evaluation.

Recommended reporting language:

The proposed literature-derived Bayesian parameters and their implementation underwent expert content review by two registered medical technologists. This review assessed relevance, clarity, and appropriateness for the stated research purpose; it did not establish clinical diagnostic accuracy or probability calibration.

12. References

1. Kurotschka PK, Gagyor I, Ebell MH. Acute Uncomplicated UTIs in Adults: Rapid Evidence Review. American Family Physician. 2024;109(2):167-174. <https://www.aafp.org/afp/2024/0200/acute-uncomplicated-utis-adults>
2. Little P, Turner S, Rumsby K, et al. Developing clinical rules to predict urinary tract infection in primary care settings. British Journal of General Practice. 2006;56(529):606-612. <https://pmc.ncbi.nlm.nih.gov/articles/PMC1874525/>
3. Bent S, Nallamotheu BK, Simel DL, Fihn SD, Saint S. Does this woman have an acute uncomplicated urinary tract infection? JAMA. 2002;287(20):2701-2710. <https://pubmed.ncbi.nlm.nih.gov/12020306/>